Estimations

1

a
- i $100 + 60$ ii 160

b
- i $60 - 20$ ii 40

c
- i 30×30 ii 900

d
- i 450×20 ii 9000

2

a
- i $900 - 100$ ii 800

b
- i $800 + 200$ ii 1000

c
- i 300×5 ii 1500

3

a
- i $7000 + 4000$ ii $11\,000$

b
- i $10\,000 - 3000$ ii 7000

4

a Many solutions including: $7000, 6360, 6400$

b Many solutions including: $3000, 2500$

c Many solutions including: $15\,000, 15\,100$

d Many solutions including: $20\,000, 22\,500$

e Many solutions including: $16\,000, 18\,000$

f Many solutions including: $400, 410$

g Many solutions including: $200, 205$

h Many solutions including: $6360, 5860, 5900$

5 **B**

6

a 1790 b 1700 c 1785 d Part (a)

7

a 740 b 700 c 743.54 d Part (a)

8

a
- i No solution provided.
- ii 127.5

b
- i No solution provided.
- ii 140.0

c
- i No solution provided.
- ii 91.3

d
- i No solution provided.
- ii 2915.4

9

a
- i No solution provided.
- ii 3.01

b
- i No solution provided.
- ii 14.6

c
- i No solution provided.
- ii 60.8

d
- i No solution provided.
- ii 30.5

Checking reasonableness

10

a No　b Yes　c No　d No
e No　f Yes

11

a Yes　b No　c Yes　d No
e Yes　f No　g Yes　h No
i Yes

12

a 80 and 2　b 82

13

a No　b Yes　c Yes　d No

14 Yes

15

a False **b** False **c** True **d** False

e True

Applications

16 $48\,000$ passengers

17 60 months

18 Yes

19

a No, because he is rounding too much. Two kebabs cost $\$19$, so he cannot afford two kebabs.

b Yes, because then they will have $\$23.40$ which will cover the cost of two kebabs.